✅ **1. Competency #7: Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations**

Highly relevant (core competency)

- The core of this problem is numerical manipulation and unit conversion:
  - Converting kilometers → meters → centimeters (multiplying by 1,000 and then by 100)
  - Performing division with large numbers: 900,000 ÷ 22,500
- This directly aligns with "manipulating numbers and information," a foundational skill in *employ.pdf*.

✅ **2. Competency #3: Implement a given strategy to determine a mathematical solution**

Highly relevant

- The solution strategy is clear and standard:
  Average step length = Total distance ÷ Total number of steps
- Students must execute this straightforward strategy, including unit alignment and computation.
- Although simple, this exemplifies "implementing an obvious or given strategy," which is precisely what this competency describes.

✅ **3. Competency #13: Use an understanding of the context to guide or expedite the mathematical solving process, e.g., working to a context-appropriate level of accuracy**

Moderately to highly relevant

- The prompt asks students to "Estimate" Toshi's average step length.
- Therefore, extreme precision (e.g., decimals beyond one place) is unnecessary; a reasonable whole number (like 40 cm) is expected.
- Additionally, real-world knowledge—that typical human step lengths range from about 40–80 cm—can help verify if the answer is plausible. This shows how context informs and validates the solution process.

✅ **4. Competency #1: Perform a simple calculation**

Moderately relevant

- The final arithmetic (900,000 ÷ 22,500 = 40) is essentially a simple division, albeit with large numbers.
- However, because it is embedded within a multi-step process involving unit conversion, it goes beyond a "bare" calculation, making this competency less central than #7 or #3.

✅ **5. Competency #8: Articulate a solution, showing and/or summarising and presenting intermediate mathematical results**

Moderately relevant (implicitly required)

- Although the question only asks for a final boxed answer (_____ cm), a complete solution typically includes:
  - Showing the unit conversion (9 km = 900,000 cm)
  - Demonstrating the division (possibly simplified: 900,000 ÷ 22,500 = 900 ÷ 22.5 = 40)
- Thus, presenting intermediate steps supports accurate and clear solution articulation.

✅ **6. Competency #12: Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation**

Moderately relevant

- The solution clearly follows multiple steps:
  1. Interpret the problem (find average step length)
  2. Convert units consistently (km → cm)
  3. Apply the formula (total distance ÷ number of steps)
  4. Compute and express the answer with correct units
- This is a classic example of a structured, multi-step procedural task.

✅ 1. Competency #7: Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations

高度相关（核心）

- 本题核心操作是数值运算与单位换算：
  - 将公里 → 米 → 厘米（涉及乘以 1,000 和 100）
  - 执行大数除法：900,000 ÷ 22,500
- 这直接对应“manipulate numbers and information”，是 *employ.pdf* 中最基础也最关键的技能。

✅ 2. Competency #3: Implement a given strategy to determine a mathematical solution

高度相关

- 解题策略明确且标准：
  平均步长 = 总距离 ÷ 总步数
- 学生需执行这一既定策略，包括单位统一和计算。
- 虽然策略简单，但“implementing a given or obvious strategy” is precisely what this competency describes.

✅ 3. Competency #13: Use an understanding of the context to guide or expedite the mathematical solving process, e.g., working to a context-appropriate level of accuracy

中高相关

- 情境提示这是“估算”（*Estimate* Toshi's average step length）。
- 因此，学生无需追求极高精度（如小数点后多位），而应给出合理整数答案（如 40 cm，而非 40.0 或 39.99…）。
- 此外，知道“人的步长大约在 40–80 cm”这一常识可帮助验证答案合理性——这体现了情境引导解题。

✅ 4. Competency #1: Perform a simple calculation

中等相关

- 最终计算（900,000 ÷ 22,500 = 40）虽涉及大数，但本质是简单除法。
- 然而，因包含单位换算，它不只是“bare calculation”，而是嵌入在更复杂的流程中，故相关性略低于 #7 和 #3。

✅ 5. Competency #8: Articulate a solution, showing and/or summarising and presenting intermediate mathematical results

中等相关（隐含要求）

- 虽然题目只要求填写最终答案（_____ cm），但在完整解答过程中，学生通常需：
  - 写出单位换算步骤
  - 展示除法过程或简化（如先约分：900,000 ÷ 22,500 = 900 ÷ 22.5 = 40）
- 因此，presenting intermediate results supports correct solution articulation.

✅ 6. Competency #12: Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation

中等相关

- 解题明确包含多个步骤：
  1. 理解问题（求平均步长）
  2. 统一单位（km → cm）
  3. 应用公式（总距离 ÷ 步数）
  4. 计算并给出带单位的答案
- 这是一个典型的多步程序性任务。